higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL INSTRUMENTS N5

(8080205)

15 April 2025 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages, 1 formula sheet and 5 graphs.

273Q1A1525

Copyright reserved

GET IT ON Google Play

Try our Apps on Play Store

NATED Question Papers	**N1-N6** Mathematics NATED	**N1-N4** Engineering Science NATED	**N1-N6** Industrial Electronics NATED	**N4-N6** Electrotechnics NATED	**N4-N6** Mechanotechnics NATED
N5-N6 Power Machines NATED	**N4-N6** Quantity Surveying NATED	**N5-N6** Strength of Materials & Structures NATED	**N1-N3** Engineering Drawing NATED	**N3** Eletro-technology NATED	**N3** Mechano-technology NATED
N4-N6 Building Administration NATED	**N4-N6** Building& Structural Surveying NATED	**N4-N6** Building & Structural Construction NATED	**N1-N3** Building Science NATED	**N1-N3** Building Drawing NATED	More…
N4-N6 Computer Practice NATED	**N4-N6** Entrepreneurship and Business Management NATED	**N4** Management Communication NATED	**N4-N6** Communication NATED	**N4-N6** Public Administration NATED	**N4-N6** Financial Accounting NATED
N4-N6 Computerised Financial Systems NATED	**N5-N6** Sales Management NATED	**N4-N6** Office Practice NATED	**N5-N6** Public Finance NATED	**N4-N6** Personnel Management NATED	**N5-N6** Municipal Administration NATED
N4-N6 Information Processing NATED	**N5-N6** Personnel Training NATED	**N4-N6** Travel Services NATED	**N6** Public Law NATED	**N5-N6** Cost and Management Accounting NATED	More…

DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL INSTRUMENTS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a black or blue pen.
5. Write neatly and legibly.

Copyright reserved

SECTION A: FLOW MEASUREMENT

QUESTION 1

1.1 Explain the difference between the rectangular, Cippoletti and V-notch open channel flowmeter. (3)

1.2 Name TWO main categories of open channel flow flowmeters. (2)

1.3 Various options are given as possible answers to the following questions. Choose the correct answer and write only the letter (A–D) next to the question number (1.3.1–1.3.4) in the ANSWER BOOK.

1.3.1 Which of the following flow-measuring elements is inherently linear and requires no signal characterisation (e.g. square-root extraction) anywhere in the loop?

A Target flowmeter
B Venturi tube
C Orifice plate
D Turbine flowmeter

1.3.2 Bernoulli's Equation is a mathematical expression of:

A The ratio of kinetic to viscous forces in a flow stream
B Friction loss as fluid moves through a rough pipe
C Potential and kinetic energies in a flow stream
D Vertical height and pressure for a static fluid

1.3.3 As an incompressible fluid moves through a restriction,

A velocity increases and pressure decreases.
B velocity decreases and pressure increases.
C velocity decreases and pressure remains the same.
D both velocity and pressure increase.

1.3.4 What type of dynamic fluid effect is illustrated by a flag flapping in the breeze?

A Transitional flow
B Laminar flow
C Vortex shedding
D Cavitation

(4 × 2) (8)

1.4 How do you identify an orifice in the pipeline? (2)

1.5 Why does flow measurement in Bernoulli primary flow device have to be rooted? (3)

[18]

QUESTION 2

Calculate the bore inside diameter of an orifice plate stainless steel pipe.

- D + D/2 taps
- Maximum flow of gas is 740 m^3/h
- Diameter of the pipe is 250 mm
- Maximum differential pressure is 1 000 Pa
- Density of the gas is 6,408 kg/m^3
- The pipeline pressure is controlled at 375 mm W.G (3,6764 Pa)
- Gas temperature is 30 ºC
- The specific heat ratio is 1,4
- Area relation is 0,32 **[12]**

TOTAL SECTION A: 30

SECTION B

QUESTION 3: DENSITY, HUMIDITY &VISCOSITY MEASUREMENT

3.1 Water content on solids will absorb IR radiation and the reflected beams will be a function of the vapour present in the solids. Make a labelled sketch of IR absorption hygrometer for solids substances. (7)

3.2 Explain the operating principle of the meter in QUESTION 3.1. (3)

3.3 A hydrometer is used to measure density of an unknown liquid by comparing it against known density, normally that of water.

3.3.1 Make a neatly labelled sketch of a hydrometer that matches the above description. (4)

3.3.2 Explain the operation of the hydrometer described in QUESTION 3.3.1. (4)

3.4 Define viscosity and name TWO factors that influence the viscosity of fluids. (3)

3.5 Make a neat labelled sketch showing the installation of a vibrating-reed viscometer in practice to ensure that temperature, pressure and flow rate are controlled. (4)

[25]

Copyright reserved

QUESTION 4: PH AND CONDUCTIVITY MEASUREMENT

4.1 Define the term pH of a solution and briefly explain how pH meter works. (3)

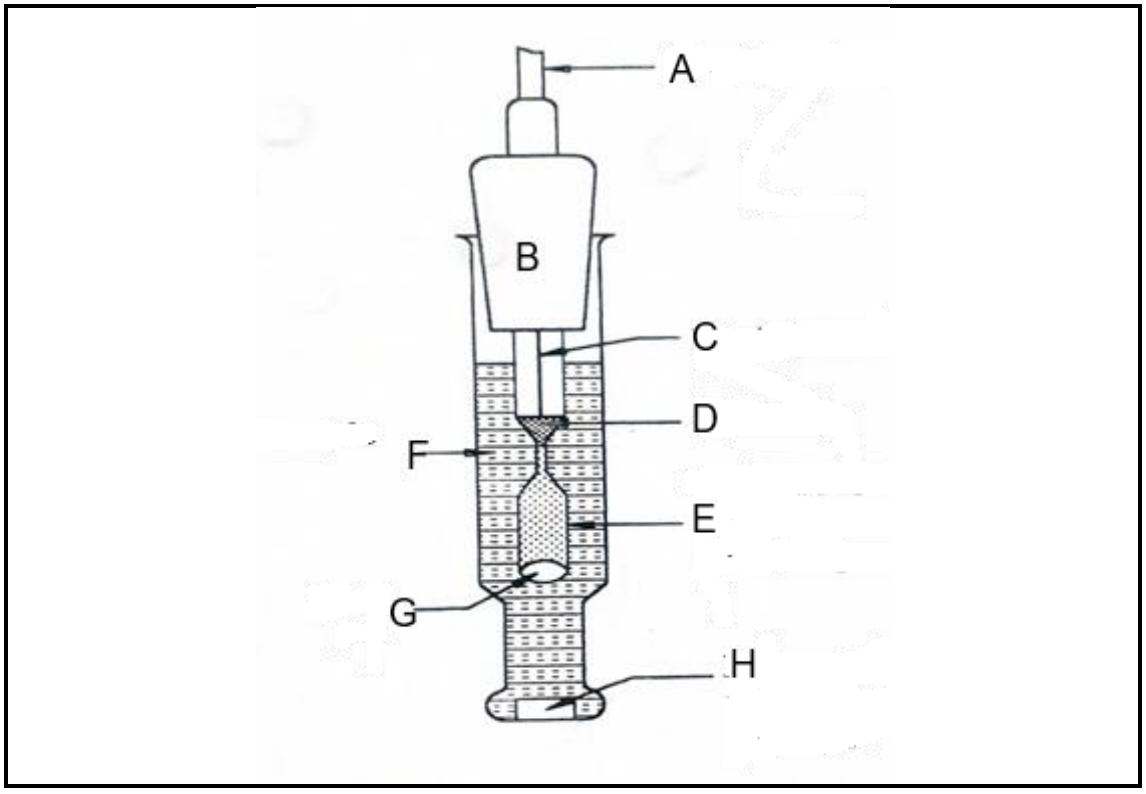
4.2 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (4.2.1–4.2.2) in the ANSWER BOOK.

4.2.1 pH is defined as the negative logarithm of hydrogen ion concentration.

4.2.2 Pure water is a weak electrolyte which dissociates to form hydrogen ions and hydroxyl ions.

(2 × 1) (2)

4.3 A typical calomel reference electrode sketch is shown below. Provide labels for the parts marked A, B, C, D, E, F, G and H.



(8)

4.4 State Faraday's law of electrolysis. (4)

4.5 Make a neatly labelled sketch of electrodeless conductivity system. (7)

[24]

QUESTION 5: AUTOMATIC CONTROL

5.1 What is the difference between controlled and manipulated variables? (3)

5.2 The most commonly used circuitry for discrimination between reference and the measured variable in electronic controllers is the operational amplifier.

5.2.1 Make a neat labelled sketch of differential amplifier or comparator as discriminating circuit in electronic controllers. (5)

5.2.2 Explain the function of the comparator circuit in QUESTION 5.2.1. (4)

5.2.3 Name three other forms of discrimination used in electronic controllers. (3)

5.3 Define the following terms:

5.3.1 Integral action time

5.3.2 Derivative action time

(2 × 2) (4)

5.4 What are the other names of integral action and derivative action? (2)
[21]

TOTAL SECTION B: 70
GRAND TOTAL: 100

Copyright reserved

(8080205)

INDUSTRIAL INSTRUMENTS N5

FORMULA SHEET

$W = 359{\cdot}2\ CZ\epsilon Ed^2\sqrt{(h\rho)}$ $\qquad R_d = W/15{\cdot}8\ \mu d$

$Q = 359{\cdot}2\ CZ\epsilon Ed^2\sqrt{(h/\rho)}$ $\qquad R_d = Q\rho/15{\cdot}8\ \mu d$

$W = 0{\cdot}01252\ CZ\epsilon Ed^2\sqrt{(h\rho)}$ $\qquad R_d = 3{\cdot}54\ W/\mu d$

$Q = 0{\cdot}01252\ CZ\epsilon Ed^2\sqrt{(h/\rho)}$ $\qquad R_d = 3{\cdot}54\ Q\rho/\mu d$

$Q_g = 2238\ CZ\epsilon Ed^2\sqrt{(h/\rho)}$ $\qquad R_d = Q_g\rho/98{\cdot}6\ \mu d$

$m = (d/D)^2$ $\qquad E = 1/\sqrt{(1 - m^2)}$

$$N = \frac{W}{0{\cdot}01252\ D^2\sqrt{(h\rho)}} = \frac{Q\sqrt{(\rho)}}{0{\cdot}01252\ D^2\sqrt{(h)}}$$

$mE = N/CZ\epsilon$ $\qquad CmE = N/Z\epsilon$ $\qquad mE = CmE/C$

$m = (d/D)^2$ $\qquad E = 1/\sqrt{(1 - m^2)}$

$$R_d = \frac{W}{15{\cdot}8\ \mu D\sqrt{(m)}} = \frac{Q\rho}{15{\cdot}8\ \mu D\sqrt{(m)}} = \frac{Q_g\rho}{98{\cdot}6\ \mu D\sqrt{(m)}}$$

$$N = \frac{W}{359{\cdot}2\ D^2\sqrt{(h)}} = \frac{Q\sqrt{(\rho)}}{359{\cdot}2\ D^2\sqrt{(h)}} = \frac{Q_g\sqrt{(\rho)}}{2238\ D^2\sqrt{(h}$$

$d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$ $\qquad mE = N/CZ\epsilon$

$W = 1890\ Ud^2\sqrt{(\rho P)}$ for critical flow/vir kritieke vloei

$CmE = N/Z\epsilon$ $\qquad mE = CmE/C$

$d = [W/1890\ U\sqrt{(\rho P)}]^{\frac{1}{2}}$ for critical flow/vir kritieke vloei

$$R_d = \frac{3{\cdot}54W}{\mu D\sqrt{(m)}} = \frac{3{\cdot}54Q\rho}{\mu D\sqrt{(m)}}$$ $\qquad d/D = [(mE)^2/1 + (mE)^2]^{\frac{1}{4}}$

$W = 1{\cdot}252\ Ud^2\sqrt{(\rho P}$ for critical flow/vir kritieke vloei

$d = [W/1{\cdot}252\ U\sqrt{(\rho P)}]^{\frac{1}{2}}$ for critical flow/vir kritieke vloei

1 kPa = 102 mmWD = 102 mmWG

1 lb/ft^3 = 16,0183 kg/m^3

Atmospheric pressure/Atmosferiese druk = 101,325 kPa

Gravitation acceleration/Gravitasieversnelling = 9,81 m/s^2

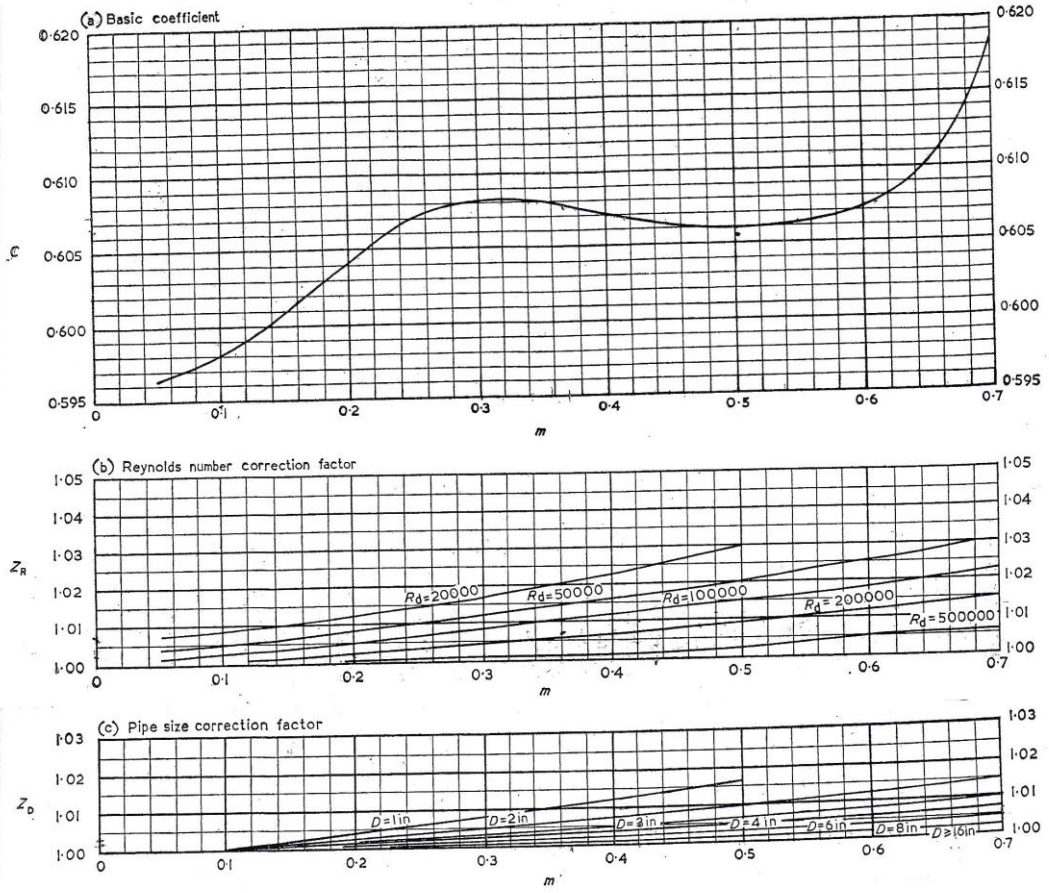
For/Vir $D + D/2$ tappings and/en flange tappings/flenstappe

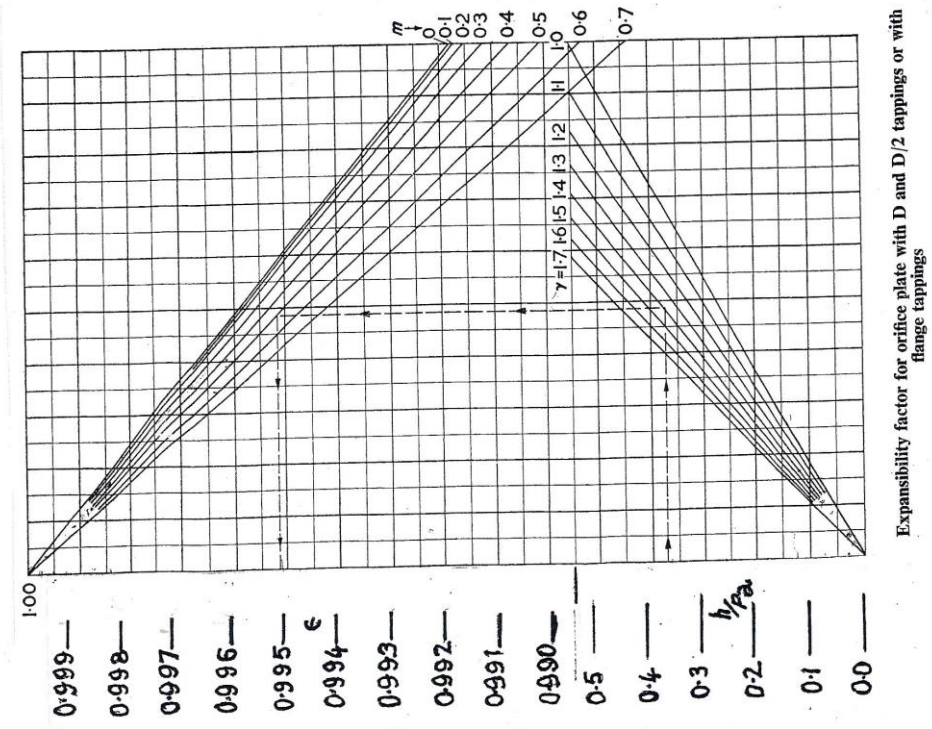
$$\frac{h}{Pa} \times 27{,}2 = \frac{kPa}{kPa} \times 27{,}2$$

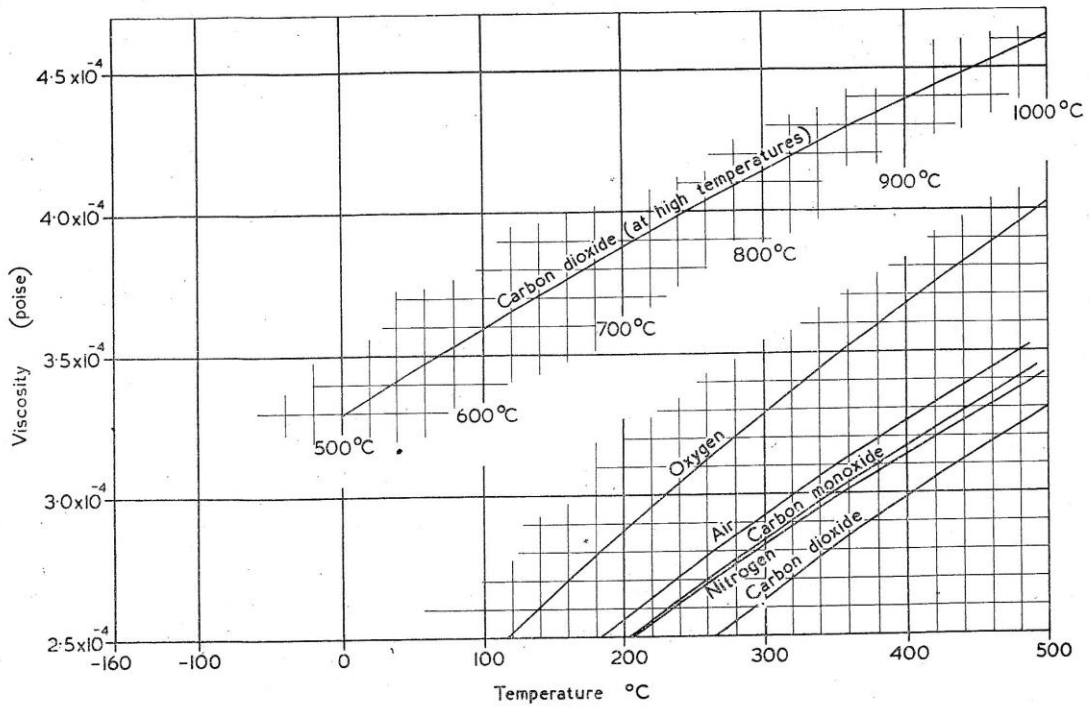
$$Q = \frac{8}{15}\ Tan\ \frac{\theta}{2}\ \sqrt{2g.H^5} \qquad Q = \frac{2}{3}\ B\ \sqrt{2g.H^3}$$

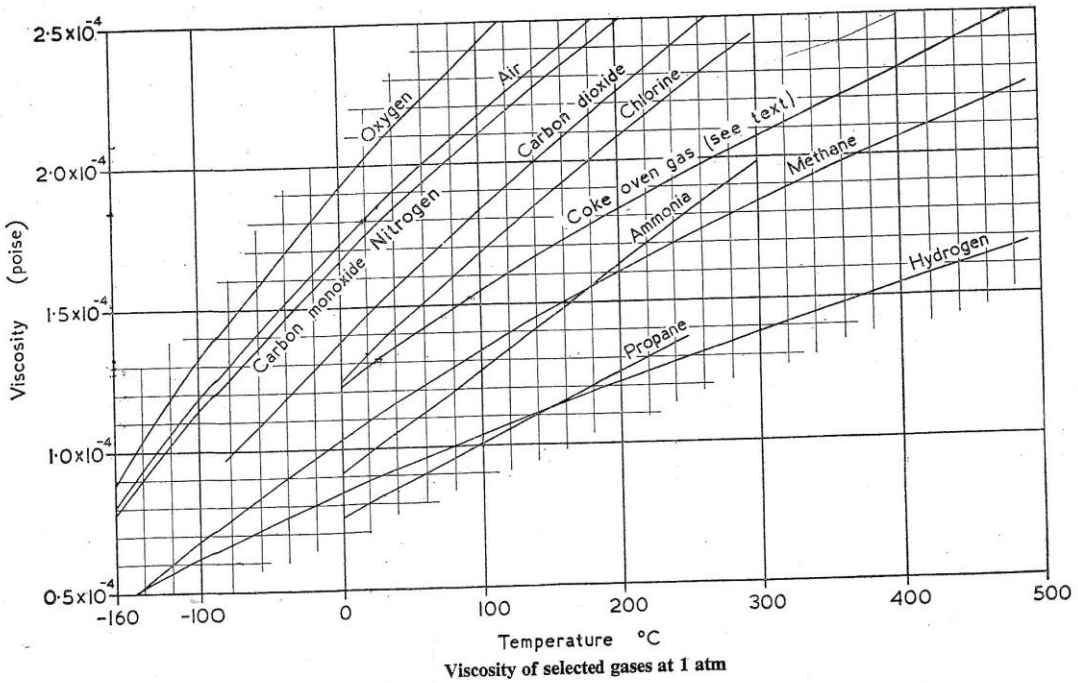
Copyright reserved

GRAPH SHEET


(a) Basic coefficient
C
m
(b) Reynolds number correction factor
Z_R
R_d=20000
R_d=50000
R_d=100000
R_d=200000
R_d=500000
(c) Pipe size correction factor
Z_D
D=1in
D=2in
D=3in
D=4in
D=6in
D=8in
D≥16in


m
γ=1·7
Y
ε
Expansibility factor for orifice plate with D and D/2 tappings or with flange tappings





Viscosity of selected gases at 1 atm

Copyright reserved

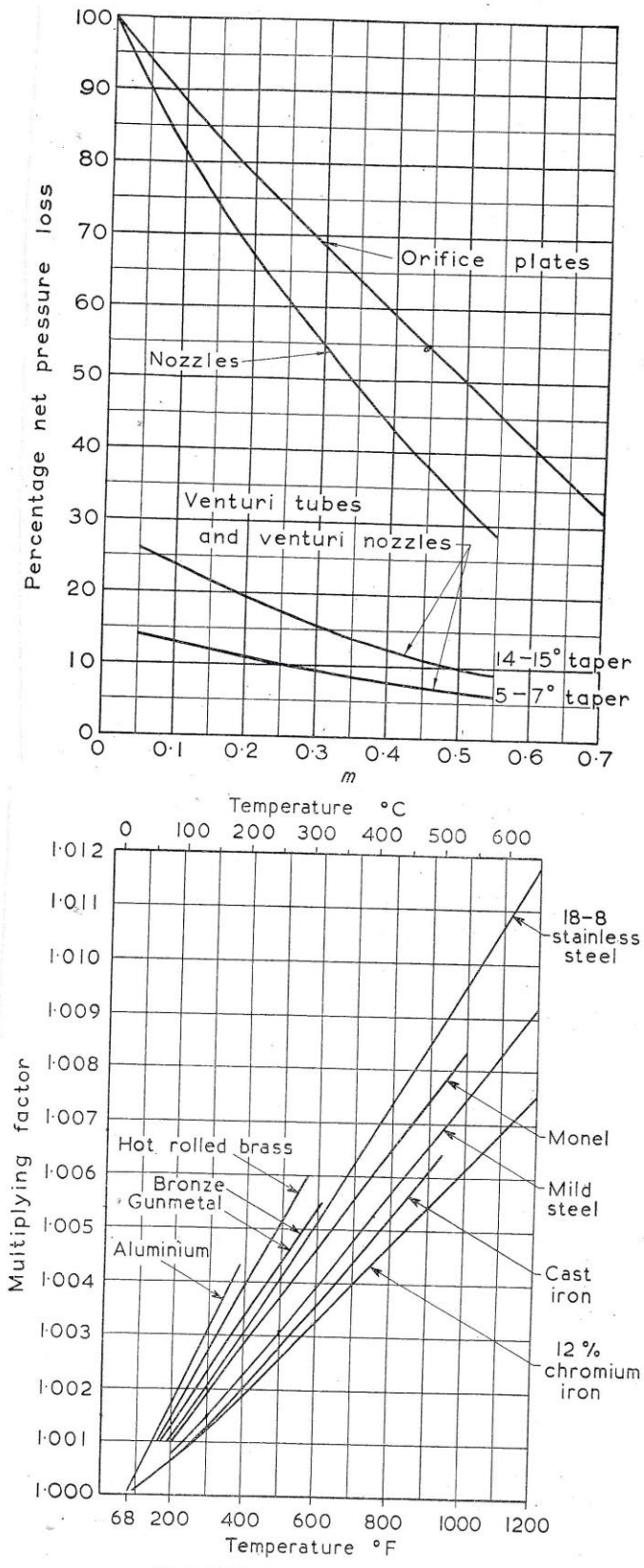
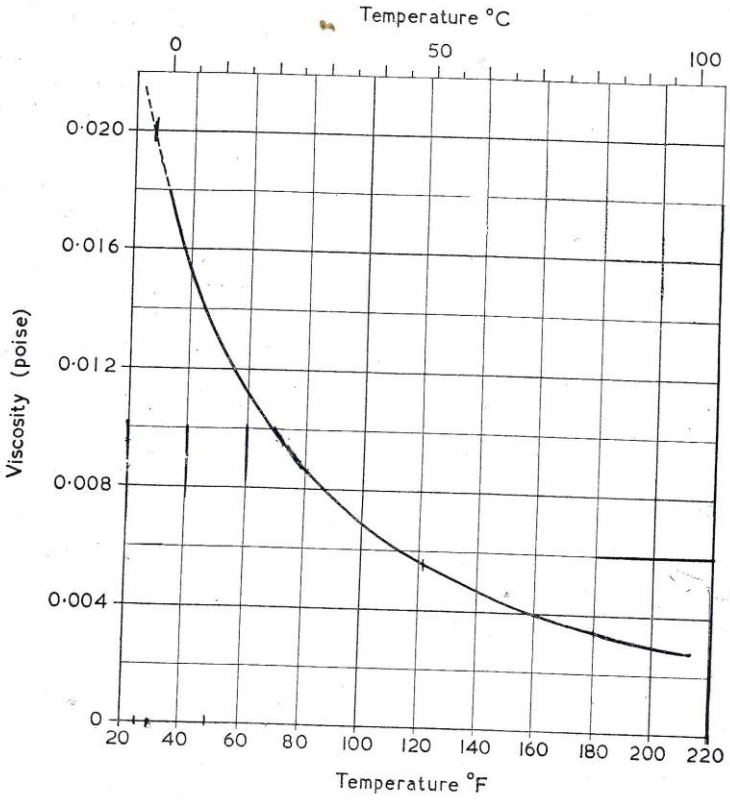
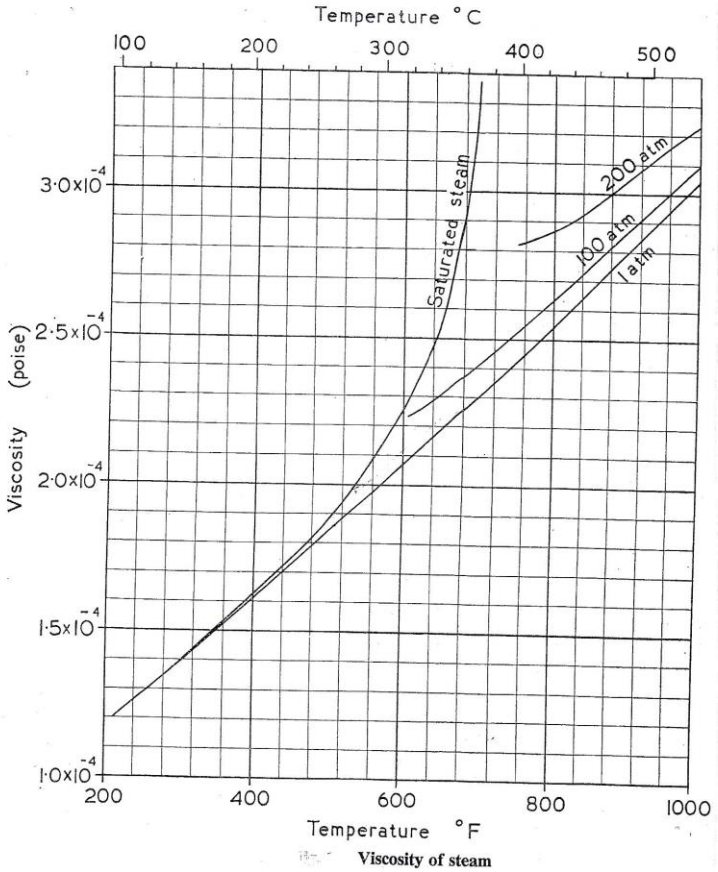


Fig. 5. Multiplying factors for thermal expansion

Copyright reserved



Viscosity of water at 1 atm

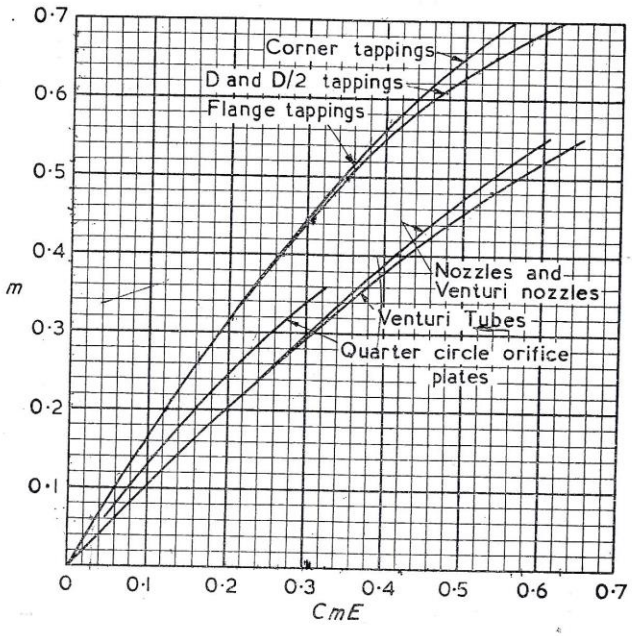


Viscosity of steam

Copyright reserved

TABLE 5. DENSITY (IN LB/FT³) OF SATURATED AND SUPERHEATED STEAM*

Absolute pressure (lbf/in²)	Saturated steam	Superheated steam at a total temperature of								
		250°F	300°F	350°F	400°F	450°F	500°F	550°F	600°F	650°F
14	0·03559	0·03351	0·03121	0·02922	0·02749	0·02595	0·02458	0·02335	0·02224	0·02123
15	0·03795	0·03593	0·03347	0·03133	0·02946	0·02781	0·02634	0·02502	0·02383	0·02275
16	0·04032	0·03834	0·03571	0·03342	0·03143	0·02966	0·02810	0·02669	0·02542	0·02427
17	0·04266	0·04080	0·03798	0·03555	0·03342	0·03154	0·02987	0·02837	0·02702	0·02579
18	0·04500	0·04322	0·04023	0·03765	0·03539	0·03339	0·03162	0·03004	0·02860	0·02731
19	0·04735	0·04566	0·04248	0·03976	0·03737	0·03526	0·03340	0·03172	0·03019	0·02883
20	0·04968	0·04810	0·04474	0·04186	0·03934	0·03713	0·03516	0·03339	0·03180	0·03034
22	0·05432	0·05299	0·04926	0·04608	0·04329	0·04085	0·03868	0·03674	0·03498	0·03338
24	0·05893	0·05790	0·05382	0·05033	0·04726	0·04458	0·04221	0·04008	0·03815	0·0364^
26	0·06353	0·06281	0·05838	0·05456	0·05123	0·04833	0·04575	0·04342	0·04136	0·0394
28	0·06812	0·06775	0·06289	0·05879	0·05522	0·05206	0·04929	0·04679	0·04456	0·0425
30	0·07273	0·07267	0·06748	0·06305	0·05917	0·05580	0·05283	0·05015	0·04776	0·04556
32	0·07722	—	0·07215	0·06739	0·06321	0·05959	0·05640	0·05353	0·05097	0·04861
34	0·08177	—	0·07675	0·07163	0·06720	0·06337	0·05995	0·05691	0·05417	0·05168
36	0·08621	—	0·08130	0·07587	0·07117	0·06711	0·06349	0·06024	0·05737	0·05473
38	0·09066	—	0·08591	0·08019	0·07519	0·07087	0·06702	0·06361	0·06057	0·05777
40	0·09515	—	0·09058	0·08446	0·07924	0·07468	0·07062	0·06698	0·06378	0·06083
42	0·09970	—	0·09524	0·08881	0·08326	0·07843	0·07413	0·07037	0·06698	0·06390
44	0·1041	—	0·09990	0·09311	0·08726	0·08217	0·07770	0·07375	0·07022	0·06693
46	0·1086	—	0·1045	0·09737	0·09132	0·08598	0·08137	0·07716	0·07342	0·07003
48	0·1130	—	0·1093	0·1017	0·09542	0·08985	0·08489	0·08052	0·07663	0·07305
50	0·1174	—	0·1140	0·1061	0·09950	0·09363	0·08849	0·08389	0·07987	0·07616
52	0·1218	—	0·1187	0·1104	0·1035	0·09737	0·09208	0·08734	0·08313	0·07924
54	0·1262	—	0·1234	0·1148	0·1075	0·1012	0·09560	0·09074	0·08636	0·08230
56	0·1305	—	0·1281	0·1192	0·1116	0·1050	0·09921	0·09416	0·08952	0·08540
58	0·1349	—	0·1329	0·1236	0·1157	0·1089	0·1029	0·09756	0·09276	0·08842
60	0·1393	—	0·1377	0·1280	0·1198	0·1127	0·1065	0·1009	0·09597	0·09149



Copyright reserved